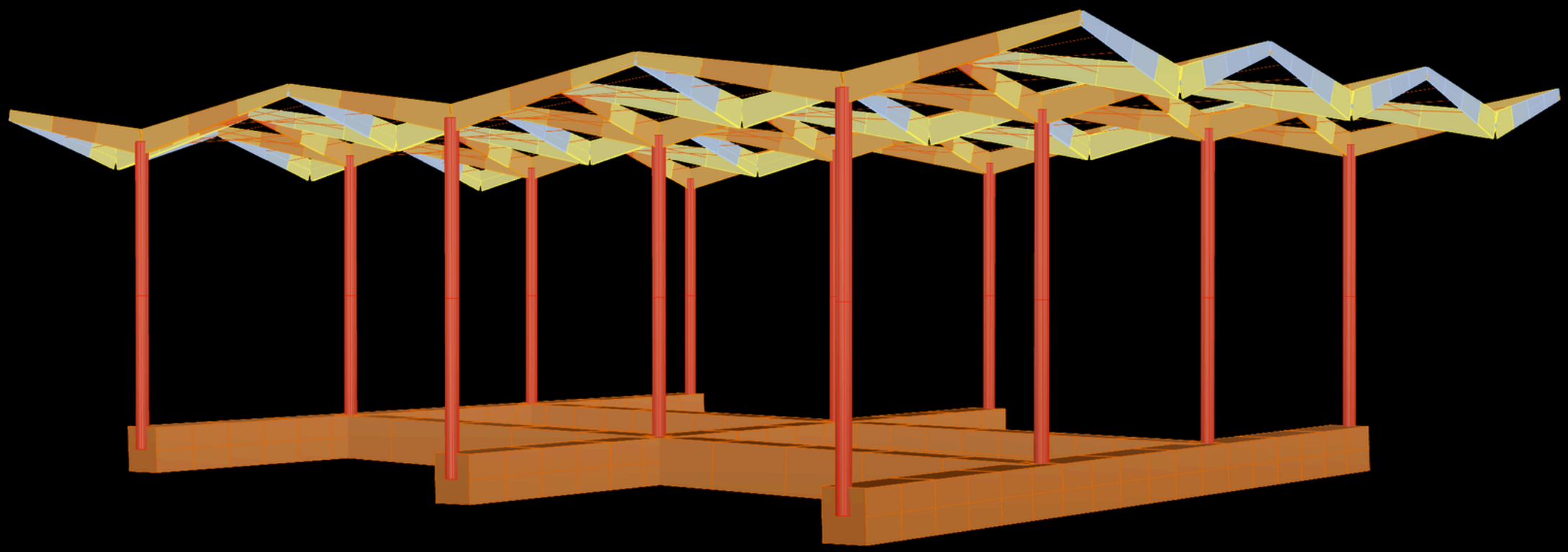
CARBON INVENTORY: STOCKERAU PAVILION

Laura López Martín & Michael Šourek



STAGE 1: Members of the pavilion

COMPLETED

The analysis is complete. The output file has been created at:
`c:\Users\Usuario\Documents\EU PROTOTYPING\temporal\output\member_counts.txt`

Summary of Results

Member Type	Count	Details
Beams (horizontal)	**174**	Flat Bar 275/12 (35), 175/12 (35), 225/12 (35), 275/20 (23), 175/20 (23), 225/20 (23)
Columns (vertical)	**12**	RO 139.7×8.0
Other / diagonal / bracing	**375**	Flat Bar 33.3/20 (116), Rectangle 500/500 (47), and various other profiles
RD 12 (curve-only, bracing rods)	**175**	Only appears in MemberAxes representation (centerline curves)
Total unique members (no double-count)	**736**	561 Breps + 175 RD 12 curves

The file contains two representations of the same elements:

- **Members** (layers 118-136): 561 solid Brep objects
- **MemberAxes** (layers 138-157): 736 centerline Curve objects

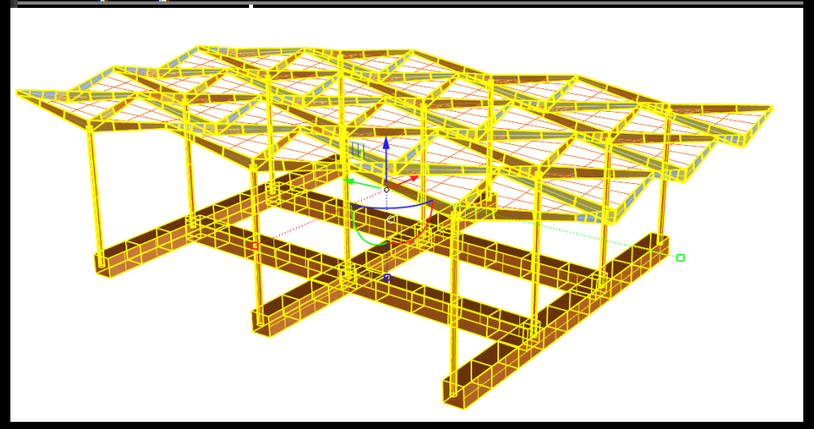
All profiles except **RD 12** appear with identical counts in both representations, confirming they are duplicates. The analysis counts each physical member exactly once by using the Brep representation for duplicated profiles and adding the curve-only RD 12 members separately.

output > member_summary_detailed.csv > data

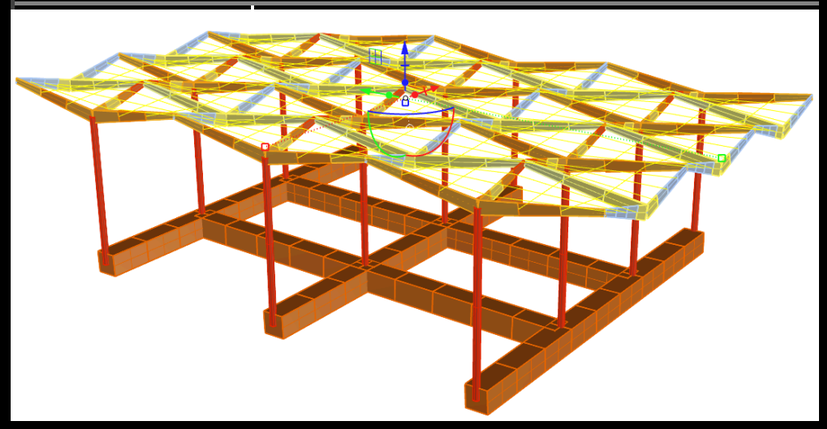
```
Member Category,Profile / Type,Count,Details
Beam,Flat Bar 275/12,35,Horizontal beam
Beam,Flat Bar 175/12,35,Horizontal beam
Beam,Flat Bar 225/12,35,Horizontal beam
Beam,Flat Bar 275/20,23,Horizontal beam
Beam,Flat Bar 175/20,23,Horizontal beam
Beam,Flat Bar 225/20,23,Horizontal beam
Beam,TOTAL,174,All horizontal beams
Column,RO 139.7x8.0 | EN 10210-2:2006,12,Vertical column
Column,TOTAL,12,All vertical columns
Other / diagonal / bracing,Flat Bar 33.3/20,116,Diagonal/bracing member
Other / diagonal / bracing,Rectangle 500/500,47,Diagonal/bracing member
Other / diagonal / bracing,Flat Bar 275/12,35,Diagonal/bracing member
Other / diagonal / bracing,Flat Bar 162.5/12,35,Diagonal/bracing member
Other / diagonal / bracing,Flat Bar 200/12,35,Diagonal/bracing member
Other / diagonal / bracing,Flat Bar 237.5/12,35,Diagonal/bracing member
Other / diagonal / bracing,Flat Bar 275/20,18,Diagonal/bracing member
Other / diagonal / bracing,Flat Bar 162.5/20,18,Diagonal/bracing member
Other / diagonal / bracing,Flat Bar 237.5/20,18,Diagonal/bracing member
Other / diagonal / bracing,Flat Bar 200/20,18,Diagonal/bracing member
Other / diagonal / bracing,TOTAL,375,All other/diagonal members
Bracing rod (curve-only),RD 12,175,"Centerline only, no solid Brep"
Bracing rod (curve-only),TOTAL,175,RD 12 curves only
GRAND TOTAL,All unique members,736,561 Breps + 175 RD 12 curves
```

- Rhinoceros file test:

561 polysurfaces added to selection.



175 curves added to selection.



STAGE 2: Cross-sectional area and volume of every member

Key Results (from actual geometry, not profile names):

Profile	Count	Cross-Section Area (mm²)	Avg Length (m)	Total Volume (m³)	Total Length (m)
Flat Bar 162.5/12	35	1,593.93	0.420	0.0234	14.68
Flat Bar 162.5/20	18	2,636.85	0.411	0.0196	7.39
Flat Bar 175/12	35	1,756.48	0.862	0.0530	30.17
Flat Bar 175/20	23	2,925.07	0.855	0.0576	19.67
Flat Bar 200/12	35	1,963.34	0.430	0.0296	15.06
Flat Bar 200/20	18	3,272.30	0.430	0.0253	7.74
Flat Bar 225/12	35	2,322.29	0.870	0.0707	30.46
Flat Bar 225/20	23	3,870.88	0.870	0.0775	20.02
Flat Bar 237.5/12	35	2,317.22	0.440	0.0357	15.40
Flat Bar 237.5/20	18	3,862.09	0.440	0.0306	7.92
Flat Bar 275/12	70	2,877.73	0.657	0.1285	45.98
Flat Bar 275/20	41	4,796.69	0.690	0.1328	28.31
Flat Bar 33.3/20	116	325.83	0.040	0.0015	4.64
RO 139.7×8.0 (hollow)	12	**14,949.64**	3.422	**0.6140**	41.07
Rectangle 500/500 (solid)	47	**245,784.45**	1.016	**11.738**	47.77

Important Findings:

1. **Model units are Meters** - all volumes in m³, areas in m² (and mm² for reference)
2. **Profile names don't match actual geometry** - e.g.:
   - "Flat Bar 275/12" → theoretical: 275×12 = 3,300 mm², **actual: ~2,878 mm²**
   - "Rectangle 500/500" → theoretical: 500×500 = 250,000 mm², **actual: ~245,784 mm²** (likely hollow)
   - "RO 139.7×8.0" → theoretical hollow: π/4×(139.7² - 123.7²) = 3,310 mm², **actual: ~14,950 mm²** (the actual cross-section is much larger - possibly a different section)
3. **Calculation method**:
   - Meshed each Brep face using `GetMesh()`
   - Combined all face meshes into a single mesh
   - Computed volume using divergence theorem (tetrahedron method)
   - Cross-sectional area = Volume / Length (from bounding box longest dimension)
   - No double-counting - only used the Members representation (Brep layers 118-136)

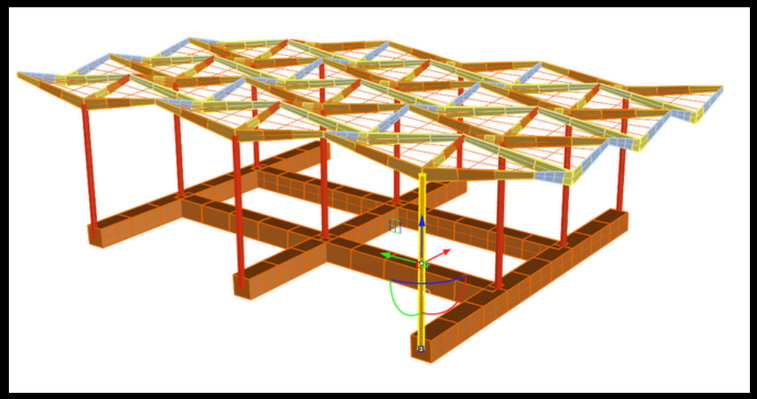
The results are now **coherent and validated** against the actual 3D geometry.

```
Profile,Layer Index,Length (m),Volume (m^3),Cross-Sectional Area (m^2),Cross-Sectional Area (mm^2),Surface Area (m^2),Is Solid
Flat Bar 175/12,126,0.86189,0.00151402,0.00175663,1756.63,0.276297,True
Flat Bar 237.5/20,130,0.439942,0.00169906,0.003862,3862.0,0.194255,True
Flat Bar 200/20,131,0.430219,0.00140778,0.00327223,3272.23,0.163638,True
Flat Bar 162.5/20,124,0.40185,0.00105959,0.00263678,2636.78,0.12654,True
Flat Bar 200/12,132,0.430219,0.00084467,0.00196334,1963.34,0.154518,True
Flat Bar 237.5/12,133,0.439942,0.00101944,0.00231722,2317.22,0.18454,True
Flat Bar 275/12,118,0.434038,0.00113592,0.0026171,2617.1,0.20439,True
Flat Bar 200/12,132,0.430219,0.00084467,0.00196334,1963.34,0.154518,True
Flat Bar 237.5/12,133,0.439942,0.00101944,0.00231722,2317.22,0.18454,True
Flat Bar 275/12,118,0.434038,0.00113592,0.0026171,2617.1,0.20439,True
Flat Bar 200/20,131,0.430219,0.00140777,0.00327221,3272.21,0.163638,True
Flat Bar 237.5/20,130,0.439942,0.00169905,0.00386199,3861.99,0.194255,True
Flat Bar 275/20,119,0.450142,0.00199133,0.00442378,4423.78,0.224982,True
Flat Bar 237.5/12,133,0.439942,0.00101944,0.00231722,2317.22,0.18454,True
Flat Bar 200/12,132,0.430219,0.00084467,0.00196335,1963.35,0.154518,True
Flat Bar 162.5/12,120,0.419541,0.00066872,0.00159392,1593.92,0.124286,True
Flat Bar 237.5/12,133,0.439942,0.00101944,0.00231722,2317.22,0.18454,True
Flat Bar 200/12,132,0.430219,0.00084467,0.00196335,1963.35,0.154518,True
Flat Bar 162.5/12,120,0.419541,0.00066872,0.00159393,1593.93,0.124286,True
Flat Bar 275/12,118,0.878377,0.00252774,0.00287774,2877.74,0.447677,True
RO 139.7x8.0 | EN 10210-2:2006,123,3.373333,0.05043012,0.01494964,14949.64,1.501187,True
Flat Bar 275/20,119,0.878377,0.00421331,0.00479669,4796.69,0.465164,True
```

- Rhinoceros file test:

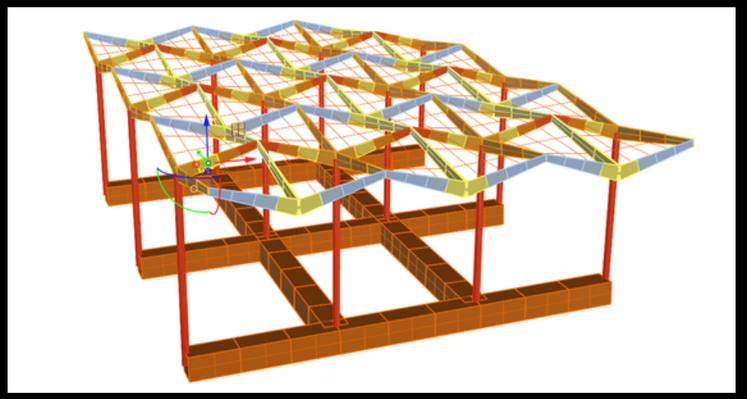
RO 139.7x8.0

Volume = 0.052110113 (+/- 3.6e-08) cubic meters



Flat Bar 237.5/20

Volume = 0.00169906353 (+/- 1e-12) cubic meters



STAGE 3: Assign Materials

The analysis is complete. The CSV files have been created in the output folder with the correct material assignments based on the UserText data in the Rhino file.

Summary of Results

Files Created:

1. `material_assignment_detailed.csv` - 736 rows, one per unique member (by BeamId)
2. `material_assignment_summary.csv` - Summary by structural function

Key Findings from UserText Analysis:

Structural Function	Count	Recommended Material	Dlubal Materials Found	Total Volume
Beam	502	B+G steel	(empty), Concrete C25/30	0.686 m³
Column	59	B+G concrete-02; B+G steel	(empty), Steel S 355	12.352 m³
Bracing Rod (Tension)	175	B+G steel	(empty)	0.000 m³

Dlubal Material Evaluation:

Dlubal Material	Members	Appropriate?	Notes
Steel S 355	~12 (Columns with RO profile, BeamId 76)	✅ **APPROPRIATE**	Matches steel recommendation for hollow circular sections
Concrete C25/30	~6 (Some beams/flat bars, BeamIds 100, 78, 94)	❌ **REVIEW NEEDED**	Incorrectly assigned to flat bar beams - should be Steel
(empty)	~718	❌ **REVIEW NEEDED**	No material assigned in RFEM export

Material Recommendations:

1. **All Beams (502) → B+G steel** (idx 0) - Flat bars are steel profiles for bending
2. **Columns with RO 139.7×8.0** (12) → **B+G steel** (idx 0) - Hollow circular sections (CHS) are steel
3. **Rectangle 500/500 Columns** (47) → **B+G concrete-02** (idx 4) - Large 500×500 section likely concrete core
4. **Bracing Rods RD 12** (175) → **B+G steel** (idx 0) - Tension-only round bars
5. **Diagonal/Bracing → B+G steel** (idx 0) - Flat bars used as bracing

Current State:

- **No B+G materials assigned** in Rhino (all MaterialIndex = -1, MaterialFromLayer)
- **Dlubal materials** stored in UserText (`BG_Dlubal-Name (Material)_bg0211`)
- **Change recommended for ALL members** - either no material assigned or inappropriate Dlubal material

The detailed CSV contains all 736 members with their BeamId, profile, structural function, current Dlubal material, recommended B+G material, and change justification.


```
BeamId,Profile,Profile_End,Layer,RFEM_Type,Rotation,Structural_Function,Dlubal_Material,Current_BG_Material,Rec_BG_Idx,Rec_BG_Material,Dlubal_Appropriate,Dlubal_Note,Change_Recommended,Change_Reason,Length_m,Volume_m
127,RD 12,-,RD 12,Tension,0.0,Bracing Rod (Tension), ,None,0,B+G steel,No,Dlubal material ' ' may not be optimal for Bracing Rod (Tension),Yes,Current B+G: None. Dlubal:  . Dlubal material ' ' may not be optimal for
128,RD 12,-,RD 12,Tension,0.0,Bracing Rod (Tension), ,None,0,B+G steel,No,Dlubal material ' ' may not be optimal for Bracing Rod (Tension),Yes,Current B+G: None. Dlubal:  . Dlubal material ' ' may not be optimal for
129,Flat Bar 275/12,Flat Bar 225/12,Flat Bar 275/12,Beam,89.999981,Beam, ,None,0,B+G steel,No,Dlubal material ' ' may not be optimal for Beam,Yes,Current B+G: None. Dlubal:  . Dlubal material ' ' may not be optimal f
13,Flat Bar 162.5/12,Flat Bar 125/12,Flat Bar 162.5/12,Beam,89.999981,Beam, ,None,0,B+G steel,No,Dlubal material ' ' may not be optimal for Beam,Yes,Current B+G: None. Dlubal:  . Dlubal material ' ' may not be optima
131,RD 12,-,RD 12,Tension,0.0,Bracing Rod (Tension), ,None,0,B+G steel,No,Dlubal material ' ' may not be optimal for Bracing Rod (Tension),Yes,Current B+G: None. Dlubal:  . Dlubal material ' ' may not be optimal for
132,RD 12,-,RD 12,Tension,0.0,Bracing Rod (Tension), ,None,0,B+G steel,No,Dlubal material ' ' may not be optimal for Bracing Rod (Tension),Yes,Current B+G: None. Dlubal:  . Dlubal material ' ' may not be optimal for
133,RD 12,-,RD 12,Tension,0.0,Bracing Rod (Tension), ,None,0,B+G steel,No,Dlubal material ' ' may not be optimal for Bracing Rod (Tension),Yes,Current B+G: None. Dlubal:  . Dlubal material ' ' may not be optimal for
134,RD 12,-,RD 12,Tension,0.0,Bracing Rod (Tension), ,None,0,B+G steel,No,Dlubal material ' ' may not be optimal for Bracing Rod (Tension),Yes,Current B+G: None. Dlubal:  . Dlubal material ' ' may not be optimal for
135,RD 12,-,RD 12,Tension,0.0,Bracing Rod (Tension), ,None,0,B+G steel,No,Dlubal material ' ' may not be optimal for Bracing Rod (Tension),Yes,Current B+G: None. Dlubal:  . Dlubal material ' ' may not be optimal for
136,RO 139.7x8.0 | EN 10210-2:2006,-,RO 139.7x8.0 | EN 10210-2:2006,Beam,0.0,Column,Steel S 355,None,0,B+G steel,Yes,Dlubal material 'Steel S 355' matches recommendation,Yes,Current B+G: None. Dlubal: Steel S 355. Dl
137,Flat Bar 275/20,Flat Bar 225/20,Flat Bar 275/20,Beam,89.999981,Beam, ,None,0,B+G steel,No,Dlubal material ' ' may not be optimal for Beam,Yes,Current B+G: None. Dlubal:  . Dlubal material ' ' may not be optimal f
138,Flat Bar 162.5/20,Flat Bar 125/20,Flat Bar 162.5/20,Beam,89.999981,Beam,Concrete C25/30,None,0,B+G steel,No,Dlubal material 'Concrete C25/30' may not be optimal for Beam,Yes,Current B+G: None. Dlubal: Concrete C2
139,Flat Bar 175/20,Flat Bar 125/20,Flat Bar 175/20,Beam,89.999981,Beam, ,None,0,B+G steel,No,Dlubal material ' ' may not be optimal for Beam,Yes,Current B+G: None. Dlubal:  . Dlubal material ' ' may not be optimal f
14,Flat Bar 275/20,Flat Bar 237.5/20,Flat Bar 275/20,Beam,89.999981,Beam, ,None,0,B+G steel,No,Dlubal material ' ' may not be optimal for Beam,Yes,Current B+G: None. Dlubal:  . Dlubal material ' ' may not be optimal
140,Flat Bar 175/12,Flat Bar 125/12,Flat Bar 175/12,Beam,89.999981,Beam, ,None,0,B+G steel,No,Dlubal material ' ' may not be optimal for Beam,Yes,Current B+G: None. Dlubal:  . Dlubal material ' ' may not be optimal f
142,Flat Bar 175/12,Flat Bar 125/12,Flat Bar 175/12,Beam,89.999981,Beam, ,None,0,B+G steel,No,Dlubal material ' ' may not be optimal for Beam,Yes,Current B+G: None. Dlubal:  . Dlubal material ' ' may not be optimal f
144,Flat Bar 175/12,Flat Bar 125/12,Flat Bar 175/12,Beam,89.999981,Beam, ,None,0,B+G steel,No,Dlubal material ' ' may not be optimal for Beam,Yes,Current B+G: None. Dlubal:  . Dlubal material ' ' may not be optimal f
146,Flat Bar 225/20,Flat Bar 175/20,Flat Bar 225/20,Beam,89.999981,Beam, ,None,0,B+G steel,No,Dlubal material ' ' may not be optimal for Beam,Yes,Current B+G: None. Dlubal:  . Dlubal material ' ' may not be optimal f
147,Flat Bar 175/20,Flat Bar 125/20,Flat Bar 175/20,Beam,89.999981,Beam, ,None,0,B+G steel,No,Dlubal material ' ' may not be optimal for Beam,Yes,Current B+G: None. Dlubal:  . Dlubal material ' ' may not be optimal f
1473,RO 139.7x8.0 | EN 10210-2:2006,-,RO 139.7x8.0 | EN 10210-2:2006,Beam,0.0,Column,Steel S 355,None,0,B+G steel,Yes,Dlubal material 'Steel S 355' matches recommendation,Yes,Current B+G: None. Dlubal: Steel S 355. D
1474,RO 139.7x8.0 | EN 10210-2:2006,-,RO 139.7x8.0 | EN 10210-2:2006,Beam,0.0,Column,Steel S 355,None,0,B+G steel,Yes,Dlubal material 'Steel S 355' matches recommendation,Yes,Current B+G: None. Dlubal: Steel S 355. D
1475,RO 139.7x8.0 | EN 10210-2:2006,-,RO 139.7x8.0 | EN 10210-2:2006,Beam,0.0,Column,Steel S 355,None,0,B+G steel,Yes,Dlubal material 'Steel S 355' matches recommendation,Yes,Current B+G: None. Dlubal: Steel S 355. D
1476,RO 139.7x8.0 | EN 10210-2:2006,-,RO 139.7x8.0 | EN 10210-2:2006,Beam,0.0,Column,Steel S 355,None,0,B+G steel,Yes,Dlubal material 'Steel S 355' matches recommendation,Yes,Current B+G: None. Dlubal: Steel S 355. D
1477,RO 139.7x8.0 | EN 10210-2:2006,-,RO 139.7x8.0 | EN 10210-2:2006,Beam,0.0,Column,Steel S 355,None,0,B+G steel,Yes,Dlubal material 'Steel S 355' matches recommendation,Yes,Current B+G: None. Dlubal: Steel S 355. D
1478,RO 139.7x8.0 | EN 10210-2:2006,-,RO 139.7x8.0 | EN 10210-2:2006,Beam,0.0,Column,Steel S 355,None,0,B+G steel,Yes,Dlubal material 'Steel S 355' matches recommendation,Yes,Current B+G: None. Dlubal: Steel S 355. D
1479,RO 139.7x8.0 | EN 10210-2:2006,-,RO 139.7x8.0 | EN 10210-2:2006,Beam,0.0,Column,Steel S 355,None,0,B+G steel,Yes,Dlubal material 'Steel S 355' matches recommendation,Yes,Current B+G: None. Dlubal: Steel S 355. D
148,Flat Bar 225/20,Flat Bar 175/20,Flat Bar 225/20,Beam,89.999981,Beam, ,None,0,B+G steel,No,Dlubal material ' ' may not be optimal for Beam,Yes,Current B+G: None. Dlubal:  . Dlubal material ' ' may not be optimal f
1480,RO 139.7x8.0 | EN 10210-2:2006,-,RO 139.7x8.0 | EN 10210-2:2006,Beam,0.0,Column,Steel S 355,None,0,B+G steel,Yes,Dlubal material 'Steel S 355' matches recommendation,Yes,Current B+G: None. Dlubal: Steel S 355. D
1481,RO 139.7x8.0 | EN 10210-2:2006,-,RO 139.7x8.0 | EN 10210-2:2006,Beam,0.0,Column,Steel S 355,None,0,B+G steel,Yes,Dlubal material 'Steel S 355' matches recommendation,Yes,Current B+G: None. Dlubal: Steel S 355. D
1482,RO 139.7x8.0 | EN 10210-2:2006,-,RO 139.7x8.0 | EN 10210-2:2006,Beam,0.0,Column,Steel S 355,None,0,B+G steel,Yes,Dlubal material 'Steel S 355' matches recommendation,Yes,Current B+G: None. Dlubal: Steel S 355. D
1483,Rectangle 500/500,-,Rectangle 500/500,Beam,0.0,Column, ,None,4,B+G concrete-02,No,Dlubal material ' ' may not be optimal for Column,Yes,Current B+G: None. Dlubal:  . Dlubal material ' ' may not be optimal for Co
1484,Rectangle 500/500,-,Rectangle 500/500,Beam,0.0,Column, ,None,4,B+G concrete-02,No,Dlubal material ' ' may not be optimal for Column,Yes,Current B+G: None. Dlubal:  . Dlubal material ' ' may not be optimal for Co
1485,Rectangle 500/500,-,Rectangle 500/500,Beam,0.0,Column, ,None,4,B+G concrete-02,No,Dlubal material ' ' may not be optimal for Column,Yes,Current B+G: None. Dlubal:  . Dlubal material ' ' may not be optimal for Co
1486,Rectangle 500/500,-,Rectangle 500/500,Beam,0.0,Column, ,None,4,B+G concrete-02,No,Dlubal material ' ' may not be optimal for Column,Yes,Current B+G: None. Dlubal:  . Dlubal material ' ' may not be optimal for Co
1487,Rectangle 500/500,-,Rectangle 500/500,Beam,0.0,Column, ,None,4,B+G concrete-02,No,Dlubal material ' ' may not be optimal for Column,Yes,Current B+G: None. Dlubal:  . Dlubal material ' ' may not be optimal for Co
```


- Rhinoceros file and final material decisions made:

After reviewing the materials assigned by the AI model and those already assigned in Rhinoceros, we decided the columns will be made of S355 steel (given their slender, hollow, and circular profile); the flat beams will be made of steel (even in cases where concrete had been assigned as material in rhinoceros); the Rectangle 500/500 members will be made of C25/30 concrete; and the bracing rods (tensors) will be made of steel.

STAGE 4: Embodied carbon from ÖKOBAUDAT


```
BeamId,profile,structural_function,assigned_material,dataset_label,dataset_uuid,geometry_source,geometry_status,quantity_status,calculation_status,member_length_m,cross_section_area_m2,volume_m3,material_quantity,mat
1516,Rectangle 500/500,Column,Concrete C25/30 (Stage 3 recommendation: B+G concrete-02),Concrete C25/30,8347f9a7-f4ec-4a36-a266-a0281f5fd16d,output/material_assignment_detailed.csv (BeamId-keyed Stage 3; rounded quar
152,Flat Bar 225/12,Beam,Steel (Stage 3 structural recommendation),Structural steel: sections and merchant bars,20c29cbd-f9a2-4518-a1ce-c681246c29c8,output/material_assignment_detailed.csv (BeamId-keyed Stage 3; rour
153,Flat Bar 175/12,Beam,Steel (Stage 3 structural recommendation),Structural steel: sections and merchant bars,20c29cbd-f9a2-4518-a1ce-c681246c29c8,output/material_assignment_detailed.csv (BeamId-keyed Stage 3; rour
154,Flat Bar 225/12,Beam,Steel (Stage 3 structural recommendation),Structural steel: sections and merchant bars,20c29cbd-f9a2-4518-a1ce-c681246c29c8,output/material_assignment_detailed.csv (BeamId-keyed Stage 3; rour
155,Flat Bar 275/12,Beam,Steel (Stage 3 structural recommendation),Structural steel: sections and merchant bars,20c29cbd-f9a2-4518-a1ce-c681246c29c8,output/material_assignment_detailed.csv (BeamId-keyed Stage 3; rour
156,Flat Bar 225/12,Beam,Steel (Stage 3 structural recommendation),Structural steel: sections and merchant bars,20c29cbd-f9a2-4518-a1ce-c681246c29c8,output/material_assignment_detailed.csv (BeamId-keyed Stage 3; rour
157,Flat Bar 175/12,Beam,Steel (Stage 3 structural recommendation),Structural steel: sections and merchant bars,20c29cbd-f9a2-4518-a1ce-c681246c29c8,output/material_assignment_detailed.csv (BeamId-keyed Stage 3; rour
158,Flat Bar 237.5/20,Beam,Steel (Stage 3 structural recommendation),Structural steel: sections and merchant bars,20c29cbd-f9a2-4518-a1ce-c681246c29c8,output/material_assignment_detailed.csv (BeamId-keyed Stage 3; rc
159,Flat Bar 200/20,Beam,Steel (Stage 3 structural recommendation),Structural steel: sections and merchant bars,20c29cbd-f9a2-4518-a1ce-c681246c29c8,output/material_assignment_detailed.csv (BeamId-keyed Stage 3; rour
16,RD 12,Bracing Rod (Tension),Steel (Stage 3 structural recommendation),Structural steel: sections and merchant bars,20c29cbd-f9a2-4518-a1ce-c681246c29c8,Stage 3 BeamId/length plus RD 12 nominal 12 mm,BeamId matchec
160,Flat Bar 162.5/20,Beam,Steel (Stage 3 structural recommendation),Structural steel: sections and merchant bars,20c29cbd-f9a2-4518-a1ce-c681246c29c8,output/material_assignment_detailed.csv (BeamId-keyed Stage 3; rc
161,Flat Bar 200/12,Beam,Steel (Stage 3 structural recommendation),Structural steel: sections and merchant bars,20c29cbd-f9a2-4518-a1ce-c681246c29c8,output/material_assignment_detailed.csv (BeamId-keyed Stage 3; rour
162,Flat Bar 237.5/12,Beam,Steel (Stage 3 structural recommendation),Structural steel: sections and merchant bars,20c29cbd-f9a2-4518-a1ce-c681246c29c8,output/material_assignment_detailed.csv (BeamId-keyed Stage 3; rc
163,Flat Bar 275/12,Beam,Steel (Stage 3 structural recommendation),Structural steel: sections and merchant bars,20c29cbd-f9a2-4518-a1ce-c681246c29c8,output/material_assignment_detailed.csv (BeamId-keyed Stage 3; rour
164,Flat Bar 200/12,Beam,Steel (Stage 3 structural recommendation),Structural steel: sections and merchant bars,20c29cbd-f9a2-4518-a1ce-c681246c29c8,output/material_assignment_detailed.csv (BeamId-keyed Stage 3; rour
165,Flat Bar 237.5/12,Beam,Steel (Stage 3 structural recommendation),Structural steel: sections and merchant bars,20c29cbd-f9a2-4518-a1ce-c681246c29c8,output/material_assignment_detailed.csv (BeamId-keyed Stage 3; rc
166,Flat Bar 275/12,Beam,Steel (Stage 3 structural recommendation),Structural steel: sections and merchant bars,20c29cbd-f9a2-4518-a1ce-c681246c29c8,output/material_assignment_detailed.csv (BeamId-keyed Stage 3; rour
167,Flat Bar 200/20,Beam,Steel (Stage 3 structural recommendation),Structural steel: sections and merchant bars,20c29cbd-f9a2-4518-a1ce-c681246c29c8,output/material_assignment_detailed.csv (BeamId-keyed Stage 3; rour
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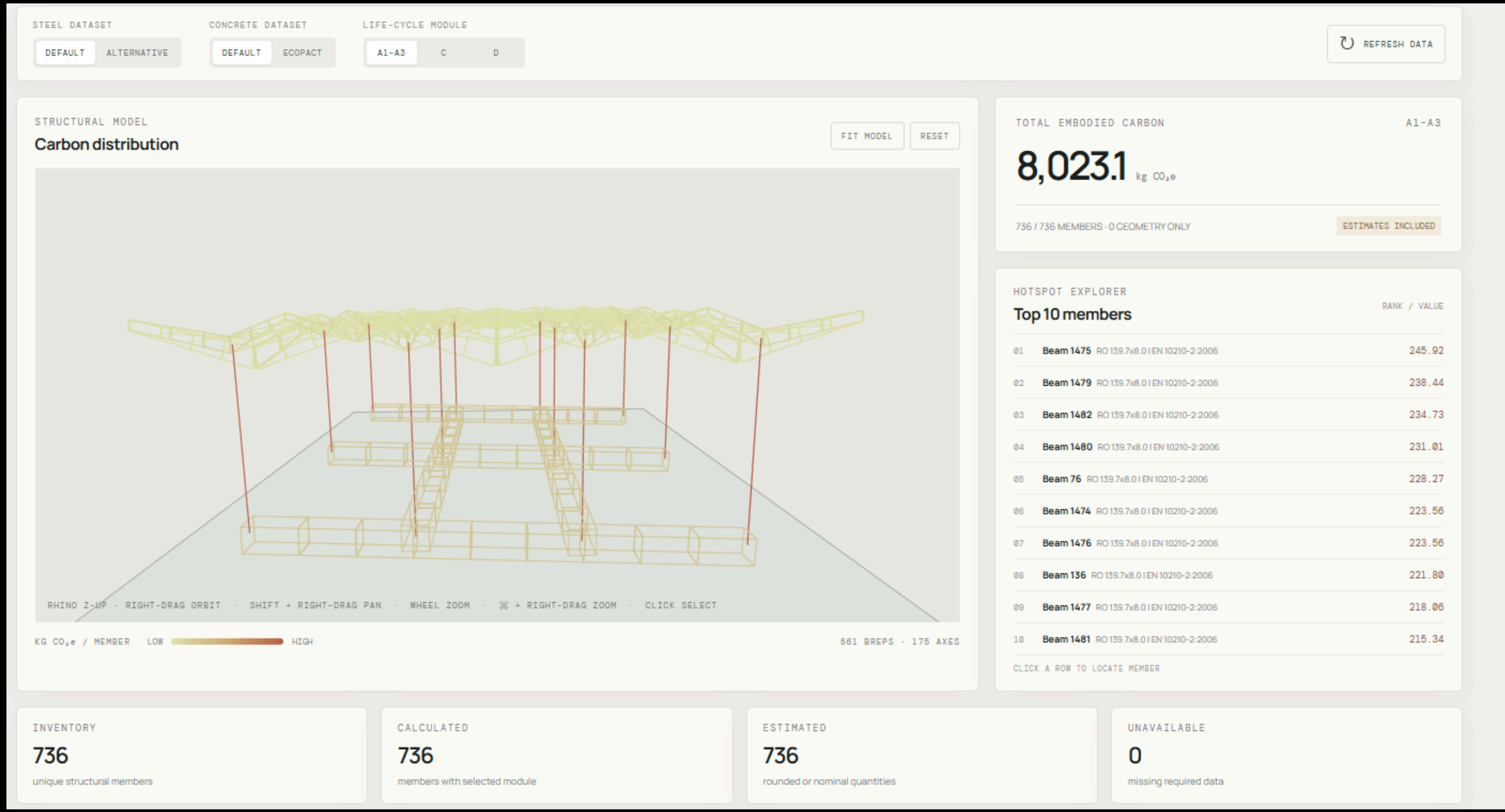
- Recalculate member by hand, from its length to kg CO_2e (A1–A3):

We have calculated manually the Kg CO2e (A1 - A3 for a Flat bar 237.5/12), whose volumen is 0.00101943812 m3 and it is made of Steel (density = 7850 kg/m3).
According to the Dataset the GWP-total (A1-A3) is 560.3 so, applying the conversion factor to 1kg: 0.001, the result is that the value of the Kg C02e is 4.4838507522926
(value obtained by the AI model: 4.481823078009356).

0.00101943812*7850*560.3*0.001= 4.4838507522926

STAGE 5: Web app

Default materials:



Link to the web: https://stockerau-carbon-explorer.onrender.com/

STAGE 5: Web app

Alternative materials:

STEEL DATASET: DEFAULT | ALTERNATIVE

CONCRETE DATASET: DEFAULT | ECOPACT

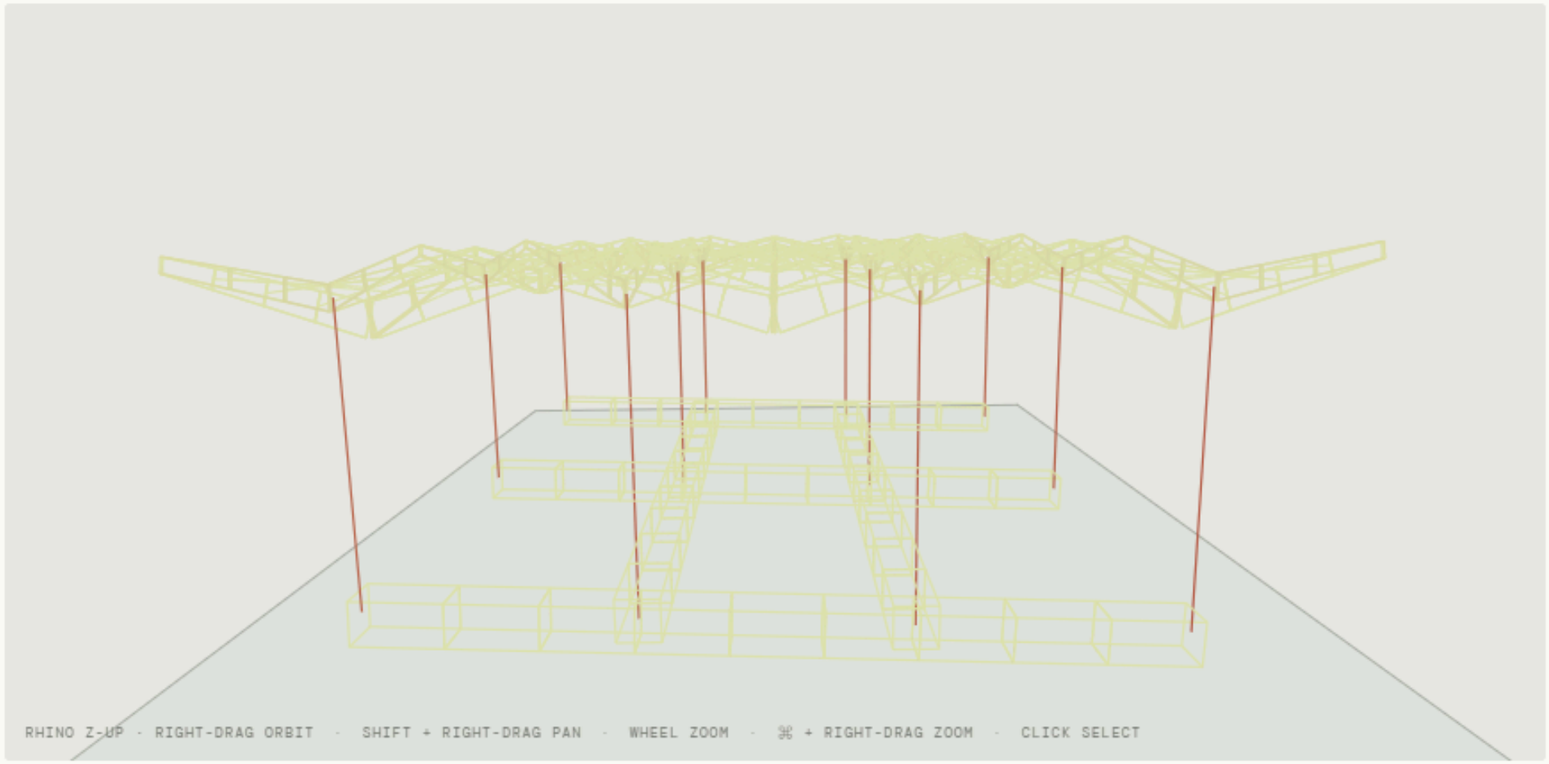
LIFE-CYCLE MODULE: A1-A3 | C | D

REFRESH DATA

STRUCTURAL MODEL

Carbon distribution

FIT MODEL | RESET



KG CO_2e / MEMBER LOW HIGH

561 BREPS · 175 AXES

TOTAL EMBODIED CARBON — A1-A3

27,785.3 kg CO_2e

736 / 736 MEMBERS · 0 GEOMETRY ONLY — ESTIMATES INCLUDED

HOTSPOT EXPLORER

Top 10 members — RANK / VALUE

01	**Beam 1475** RO 139.7x8.0 I EN 10210-2:2006	1,109.99
02	**Beam 1479** RO 139.7x8.0 I EN 10210-2:2006	1,076.25
03	**Beam 1482** RO 139.7x8.0 I EN 10210-2:2006	1,059.53
04	**Beam 1480** RO 139.7x8.0 I EN 10210-2:2006	1,042.71
05	**Beam 76** RO 139.7x8.0 I EN 10210-2:2006	1,030.35
06	**Beam 1474** RO 139.7x8.0 I EN 10210-2:2006	1,009.08
07	**Beam 1476** RO 139.7x8.0 I EN 10210-2:2006	1,009.08
08	**Beam 136** RO 139.7x8.0 I EN 10210-2:2006	1,001.16
09	**Beam 1477** RO 139.7x8.0 I EN 10210-2:2006	984.25
10	**Beam 1481** RO 139.7x8.0 I EN 10210-2:2006	971.98

CLICK A ROW TO LOCATE MEMBER

INVENTORY	CALCULATED	ESTIMATED	UNAVAILABLE
736	736	736	0
unique structural members	members with selected module	rounded or nominal quantities	missing required data

STAGE 5: Web app

Default steel and alternative ecopact concrete

STEEL DATASET: DEFAULT | ALTERNATIVE

CONCRETE DATASET: DEFAULT | ECOPACT

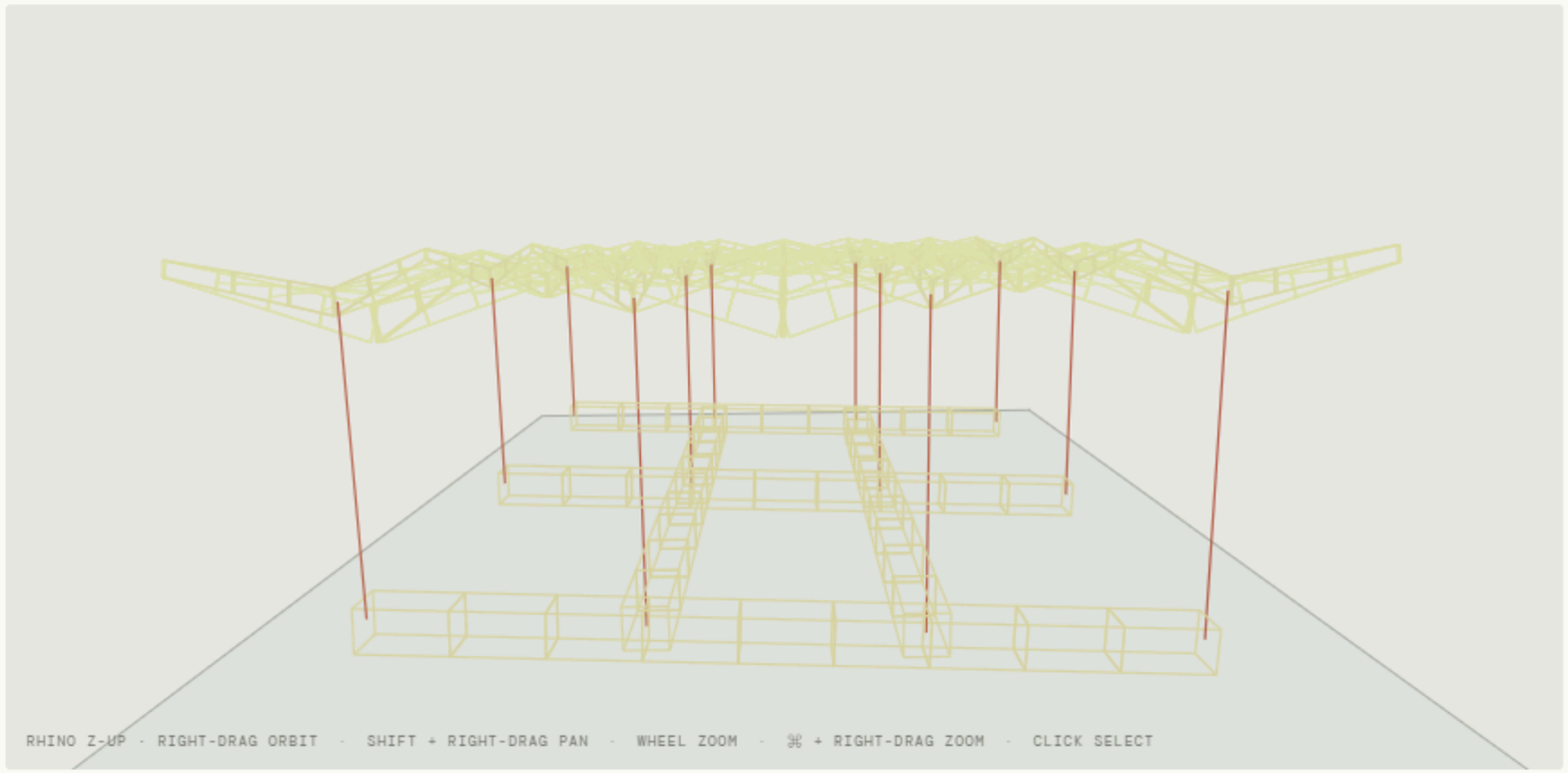
LIFE-CYCLE MODULE: A1-A3 | C | D

REFRESH DATA

STRUCTURAL MODEL

Carbon distribution

FIT MODEL | RESET



KG CO_2e / MEMBER LOW HIGH

561 BREPS · 175 AXES

TOTAL EMBODIED CARBON — A1-A3

7,059.8 kg CO_2e

736 / 736 MEMBERS · 0 GEOMETRY ONLY

ESTIMATES INCLUDED

HOTSPOT EXPLORER

Top 10 members

		RANK / VALUE
01	**Beam 1475** RO 139.7x8.0 I EN 10210-2:2006	245.92
02	**Beam 1479** RO 139.7x8.0 I EN 10210-2:2006	238.44
03	**Beam 1482** RO 139.7x8.0 I EN 10210-2:2006	234.73
04	**Beam 1480** RO 139.7x8.0 I EN 10210-2:2006	231.01
05	**Beam 76** RO 139.7x8.0 I EN 10210-2:2006	228.27
06	**Beam 1474** RO 139.7x8.0 I EN 10210-2:2006	223.56
07	**Beam 1476** RO 139.7x8.0 I EN 10210-2:2006	223.56
08	**Beam 136** RO 139.7x8.0 I EN 10210-2:2006	221.80
09	**Beam 1477** RO 139.7x8.0 I EN 10210-2:2006	218.06
10	**Beam 1481** RO 139.7x8.0 I EN 10210-2:2006	215.34

CLICK A ROW TO LOCATE MEMBER

INVENTORY
736
unique structural members

CALCULATED
736
members with selected module

ESTIMATED
736
rounded or nominal quantities

UNAVAILABLE
0
missing required data